

Information Security Project Work

Design and Implementation of an Organizational Information Security Framework

Project Description

You have been contracted as a Junior Information Security Consultant for a mid-sized organization. The organization, such as a school, hospital, bank, or logistics firm recently suffered a security incident. A phishing email tricked an employee into revealing login credentials, leading to unauthorized access to sensitive data. While the breach was contained, the management realized that they lack proper security policies, user training, and technical controls. **Your task is to conduct a comprehensive security review and design a realistic, cost-effective, and legally compliant information security framework tailored to the organization's needs.** This includes performing a security audit, recommending appropriate tools, drafting policies, and planning user awareness programs. This project is designed to simulate the kind of work information security professionals do in real organizations. Your recommendations must be practical, justified, and aligned with industry best practices and legal requirements.

Deadline: 21/07/2026

Skills To Be Gain

- Risk assessment and vulnerability analysis
- Security policy development
- Legal and ethical analysis in security
- Evaluation and selection of security tools
- User training and communication
- Hands-on security implementation (optional)

Choice of Project Context:

You are expected to choose **one** of the following approaches for this project:

- **Option 1: Real Organization** – Identify and work with an actual organization (e.g., local school, business, NGO, or hospital). Conduct informal interviews, observations, or questionnaires to gather real data (with consent).
- **Option 2: Simulated Case Study** – Use a detailed fictional organization provided by your instructor. You will assume realistic roles and work with simulated data and background information.

Instructions:

In your project, begin by writing a **Case Brief** that provides a clear overview of the organization. This should include:

1. **Name of the organization** (real or simulated)
2. **Type of organization** (e.g., school, hospital, financial institution)

3. **Size and structure** (number of staff, departments, branches)
4. **IT infrastructure overview** (networks, devices, servers, cloud services, etc.)
5. **Nature of the security issue or concern** (e.g., phishing, poor access control, lack of training)
6. **Why the organization needs an information security framework**

Project Objectives

1. **Conduct a Security Audit and Risk Assessment**
 - Identify potential threats and vulnerabilities in the organization's systems and processes.
 - Analyze roles and responsibilities of users and IT managers regarding information security.
2. **Assess Legal and Ethical Responsibilities**
 - Investigate applicable legal frameworks (e.g., Ghana Data Protection Act, GDPR).
 - Identify and discuss ethical challenges such as privacy, monitoring, and access control.
3. **Develop an Organizational Security Policy**
 - Create a policy document that includes:
 - Acceptable use
 - Password/authentication guidelines
 - Access control
 - Data handling
 - Incident reporting
4. **Recommend and Justify Security Tools**
 - Select appropriate tools (e.g., firewall, antivirus, encryption, MFA).
 - Explain how these tools address the risks identified in the audit.
5. **Design a User Awareness and Training Program**
 - Develop a training plan to educate users and managers about security best practices.
 - Include sample materials or outlines for training sessions.
6. **(Optional) Prototype or Simulate Implementation**
 - Configure or simulate security tools in a lab environment using software such as pfSense, Wireshark, or Snort.
 - Document your setup and justify your configurations.

Expected Deliverables

1. Security Audit and Risk Assessment Report
2. Legal and Ethical Compliance Brief
3. Information Security Policy Document
4. Security Tools Recommendation Report
5. User Training and Awareness Plan
6. (Optional) Prototype/Security Tool Simulation Report

Requirement – Proof of Original Work

At the end of your project report, you must include a signed Self-Declaration Statement confirming that the work is your own and not generated by AI or copied from other sources. This is **mandatory** and part of your project assessment.

Without this declaration, your project will not be accepted.

Self-Declaration Statement on AI Usage and Academic Integrity

I,, hereby declare that this project titled:
“Design and Implementation of an Organizational Information Security Framework”
is entirely my own original work.

I affirm that:

- I have **not used AI tools** (e.g., ChatGPT, Bard, or others) to generate full sections of this report.
- Any use of AI or digital tools was strictly limited to **grammar checks, formatting help, or idea prompts**, and **did not replace my personal understanding or effort**.
- I fully understand all the content presented in this project and can explain it independently if required.
- I have **not copied**, paraphrased, or submitted any material that I do not fully comprehend.

I understand that any false declaration, plagiarism, or excessive reliance on AI tools may result in **disqualification, failure, or disciplinary action**.

Student's Name: _____

Student ID: _____

Signature: _____

Date: _____